

Gautam Kumar

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PROFESSIONAL SUMMARY

Full Stack Developer with project experience across React, Node.js, FastAPI, and cloud-integrated backends. Built and shipped five end-to-end applications covering real-time AI pipelines, REST APIs, and database-driven web platforms. Comfortable working across the stack – from responsive React frontends to Python/Node.js backends and Supabase/Firestore databases. Seeking a full-stack internship or entry-level role to contribute to production systems and deepen engineering fundamentals.

EDUCATION

Bachelor of Technology (B.Tech) – Computer Science & Engineering

2023 – 2027

Gandhi Institute Of Technology (GITAM) University, Bengaluru

CGPA: 7.36

EXPERIENCE

Virtual Intern – AI & Data Analytics (Green Skills)

Oct 2025 – Nov 2025

AICTE · Shell India Markets Pvt. Ltd. · Edunet Foundation

- Completed 4-week structured program covering AI fundamentals, data-driven decision-making, and sustainability-oriented technology use cases.
- Worked through hands-on Python-based data analysis modules and industry-aligned assignments under the Skills4Future program.
- Delivered a final project applying learned concepts to a real-world green-technology scenario.

PROJECTS

AI Meeting & Lecture Note Taker

Tech Stack: React, TypeScript, Node.js, Express.js, OpenAI API, Whisper, Supabase

- Built a full-stack web app that records audio, transcribes speech via OpenAI Whisper, and generates structured summaries using the OpenAI API; end-to-end latency from upload to summary is under 30s for a 10-minute recording.
- Implemented keyword extraction and a searchable note archive backed by Supabase with real-time status updates during processing.
- Designed a React + TypeScript frontend with component-based architecture and a RESTful Express.js API layer.

Generative AI – Wildlife Conservation & Species Identification

Tech Stack: React (Vite), FastAPI, Node.js, YOLOv8, ResNet50, EfficientNet, Firestore, Cloudinary

- Trained an EfficientNet classifier on a 12-class custom wildlife dataset; achieved approximately 78 % top-1 accuracy on the held-out test set.
- Built a FastAPI backend serving model inference, automated report generation, and a Firestore-backed population monitoring dashboard.
- Developed a React (Vite) frontend with Cloudinary image uploads and a live research analytics panel.

Helmet Violation Detection System

Tech Stack: Python, OpenCV, YOLOv8, Tesseract OCR

- Built a real-time computer vision pipeline detecting helmetless motorcycle riders from video feed using YOLOv8; achieved approximately 82 % detection precision on a 500-frame test set.
- Integrated Tesseract OCR for automatic number-plate extraction, reducing manual logging effort for traffic operators.

AI-Based Plant Disease Detection System

Tech Stack: Python, TensorFlow, Keras, Flask

- Trained a CNN to classify 15 plant disease categories from leaf images; reached approximately 85 % validation accuracy on the PlantVillage dataset subset used.
- Wrapped the model in a Flask REST API for browser-based image upload and instant diagnosis output.

CrowdScope – Crowd Monitoring & Management System

Tech Stack: Python, OpenCV, Machine Learning

- Built a real-time crowd density estimator using OpenCV frame analysis and a regression model trained on simulated density data.
- System outputs density heatmaps and threshold alerts to support operational safety decisions in high-footfall scenarios.

TECHNICAL SKILLS

- **Frontend:** React, JavaScript (ES6+), TypeScript, HTML5, CSS3
- **Backend:** Node.js, Express.js, FastAPI, Flask, REST API Design
- **Databases:** MySQL, Supabase (PostgreSQL), Firebase / Firestore
- **Languages:** Python, JavaScript, TypeScript, Java, C
- **AI / ML:** TensorFlow, Keras, OpenCV, YOLOv8, ResNet50, EfficientNet
- **Cloud & Infra:** AWS (Basics), Linux (Ubuntu), Virtual Machines, Uptime Monitoring
- **Networking:** HTTP/HTTPS, REST, Client-Server Architecture, Basic Networking
- **Developer Tools:** Git, GitHub, Canva, DaVinci Resolve, AutoCAD

ACHIEVEMENTS & ACTIVITIES

- **National Cadet Corps (NCC) Cadet** – Developed leadership and structured team coordination through formal training programmes.
- **Video Editing & Creative Media** – Proficient in DaVinci Resolve; produced project demo videos and visual content.

LANGUAGES

English | Hindi | Bhojpuri | Nepali